Problem Statement — SmartRoom: occupancy-aware classroom energy control

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Context and problem

UBB uses a large number of classrooms, seminar rooms and laboratories spread over many buildings. Their lighting and heating are controlled manually: lights stay on after the last class of the day, radiators heat rooms that are empty between seminars or over the weekend, and nobody knows which rooms waste the most energy, because consumption is measured, at best, per building. This waste works against SDG 7 (target 7.3, energy efficiency) and against UBB's own Climate Action Plan, which asks for more smart buildings, motion sensors in common spaces and lower electricity use per person. It is also one of the reasons UBB loses points in the *Energy & Climate Change* criterion of UI GreenMetric.

Goal

Design an embedded system, **SmartRoom**, in which every classroom is managed by a small **Room Controller** that detects whether the room is occupied and cooperates with a central **Building Energy Manager** to keep lights and heating on only when the room is used or about to be used. The system must reduce wasted energy without reducing comfort, and must report consumption per room.

Actors

Actor	Role
Room Controller (RC)	Embedded device in each room: presence sensor, light relay, radiator valve and energy meter. Holds the room state: *Eco*, *Preheating*, *Active*, *Vacating*.
Building Energy Manager (BEM)	Central controller of a building. Knows the timetable, decides when rooms are switched on or off, and aggregates energy data.
Room users (students, teachers)	Occupy rooms; can press a manual override button on the RC.
Facility administrator	Receives reports and alerts from the BEM; sets comfort and economy setpoints.
UBB timetable system	Provides the BEM with the schedule of classes for each room.

Communications (signals)

The core of the system is the dialogue between the two main actors, **RC** and **BEM**:

Direction	Signal	Meaning
BEM ⟶ RC	`preheat`	A class is scheduled in this room in 30 minutes: bring heating to comfort level.
BEM ⟶ RC	`activate`	Room may be used: lights may be switched on by presence, heating stays at comfort.
BEM ⟶ RC	`deactivate`	Room is empty and no class follows soon: lights off, heating to economy.
RC ⟶ BEM	`occupied`	The presence sensor has detected people in the room.
RC ⟶ BEM	`empty`	No presence detected for a grace period (e.g. 10 minutes).
RC ⟶ BEM	`energyReport`	Periodic report of the room's consumption since the last report.

Secondary communications: the timetable system sends the schedule to the BEM; the BEM sends daily reports and anomaly alerts to the facility administrator (e.g. a room that consumes energy while reported empty); a user's override button makes the RC send `occupied` immediately.

Typical interaction

The room starts in *Eco*. Thirty minutes before a seminar the BEM sends `preheat`; the RC enters *Preheating*. Students arrive, the RC detects them and sends `occupied`; the BEM answers `activate` and the room is *Active*. After the seminar the RC sees no one for the grace period, enters *Vacating* and sends `empty`; if no class follows, the BEM sends `deactivate` and the room returns to *Eco*. At the end of each period the RC sends `energyReport`, which the BEM aggregates for the administrator.

Requirements and constraints

- **Safety and comfort:** a room that is occupied is never left dark or at economy temperature; the manual override always works, even if the BEM is unreachable.
- **Efficiency:** lights are never on in a room reported empty longer than the grace period.
- **Privacy:** only presence (yes/no) is sensed — no cameras, no identification of people.
- **Low cost and retrofittable**, so it can scale to all UBB buildings and count toward the smart-building targets of the Climate Action Plan.